

J1 5V INPUT 1715721	1:+5V 2:GND
F1 1.1A PTC MF-MSMF110-2	1:+5V_RAW 2:+5V_FUSED
D1 SS34 SS34	1:+5V_FUSED 2:+5V
D2 SMBJ5.0A SMBJ5.0A	1:+5V 2:GND
U6 AP2112K-3.3 AP2112K-3.3TRG1	1:GND 2:+5V 3:+5V 4:NC 5:+3V3
C1 100uF EEE-FK1A101P	1:+5V 2:GND
C2 10uF GRM21BR61A106KE19L	1:+5V 2:GND
C3 10uF GRM21BR61A106KE19L	1:+3V3 2:GND
U1 ESP32-WROOM-32E ESP32-WROOM-32E-N4	1:GND 2:+3V3 3:ESP_EN 11:RELAY_CTRL 12:RS485_DE 25:ESP_BOOT 26:ACCEL_INT 27:UART_RX 28:UART_TX 29:ACCEL_CS 30:SPI_SCK 31:SPI_MISO 33:I2C_SDA 36:I2C_SCL 37:SPI_MOS
C4 22uF GRM31CR61A226ME15L	1:+3V3 2:GND
C5 100nF GRM188R71C104KA01D	1:+3V3 2:GND
R1 10k RC0603FR-0710KL	1:+3V3 2:ESP_EN
R2 10k RC0603FR-0710KL	1:+3V3 2:ESP_BOOT
U2 TMP117M TMP117MAIDRVR	1:I2C_SCL 2:GND 3:ALERT_NC 4:I2C_SDA 5:ADD0_GND 6:+3V3

C6 100nF GRM188R71C104KA01D	1:+3V3 2:GND
U3 INA219A INA219AIDCNR	1:I2C_SCL 2:GND 3:I2C_SDA 4:+3V3 5:LOAD_IN 6:LOAD_OUT 7:GND 8:GND
R3 0.05R 1% WSL2512R0500FEA	1:LOAD_IN 2:LOAD_OUT
C7 100nF GRM188R71C104KA01D	1:+3V3 2:GND
J4 CURRENT PATH 1715721	1:LOAD_IN 2:LOAD_OUT
U4 ADXL345 ADXL345BCCZ-RL7	1:+3V3 2:+3V3 3:GND 4:GND 5:GND 6:GND 7:ACCEL_CS 8:ACCEL_INT 9:NC 10:GND 11: :SPI_MOSI 12:SPI_MISO 13:SPI_SCK 14:+3V3
C8 100nF GRM188R71C104KA01D	1:+3V3 2:GND
C9 10uF GRM21BR61A106KE19L	1:+3V3 2:GND
U5 MAX3485 MAX3485ESA+	1:UART_RX 2:RS485_DE 3:RS485_DE 4:UART_TX 5:GND 6:RS485_A 7:RS485_B 8:+3V3
C10 100nF GRM188R71C104KA01D	1:+3V3 2:GND
D3 SM712 SM712-02HTG	1:RS485_A 2:GND 3:RS485_B
R4 120R RC0805FR-07120RL	1:RS485_A_TERM 2:RS485_B_TERM
JP1 TERM ENABLE M20-9990346	1:RS485_A 2:RS485_A_TERM
JP2 BIAS A M20-9990346	1:+3V3 2:RS485_A

JP3 BIAS B M20-9990346	1:RS485_B 2:GND
R5 680R RC0603FR-07680RL	1:+3V3 2:JP2_1
R6 680R RC0603FR-07680RL	1:JP3_2 2:GND
J2 MODBUS A/B/GND 1729128	1:RS485_A 2:RS485_B 3:GND
Q1 AO3400A AO3400A	1:RELAY_GATE 2:GND 3:RELAY_COIL_LOW
R7 100R RC0603FR-07100RL	1:RELAY_CTRL 2:RELAY_GATE
R8 100k RC0603FR-07100KL	1:RELAY_GATE 2:GND
D4 SS14 SS14	1:RELAY_COIL_LOW 2:+5V
K1 G5V-1-DC5 G5V-1-DC5	1:+5V 2:RELAY_COIL_LOW 3:RELAY_COM 4:RELAY_NO
J3 RELAY COM/NO 1715721	1:RELAY_COM 2:RELAY_NO
J5 UART PROGRAM M20-9990546	1:+3V3 2:GND 3:UART_TX 4:UART_RX 5:ESP_BOOT
TP1 3V3 5000	1:+3V3
TP2 GND 5000	1:GND
TP3 SDA 5000	1:I2C_SDA

TP4 SCL
5000

1:I2C_SCL

TP5 RS485_A
5000

1:RS485_A

TP6 RS485_B
5000

1:RS485_B